

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Inverse Cosine Rational Integral

Evaluate the definite integral:

$$\int_0^{\infty} e^{\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)} \frac{2x}{(1+x^2)^2} dx$$

Solution: We begin by applying the trigonometric substitution $x = \tan(t)$. The limits of integration change from $x \in (0, \infty)$ to $t \in (0, \pi/2)$. The differential is $dx = \sec^2(t) dt$. Using the double angle identity for cosine, we simplify the argument of the inverse cosine:

$$\frac{1-x^2}{1+x^2} = \frac{1-\tan^2(t)}{1+\tan^2(t)} = \cos(2t)$$

Since $t \in (0, \pi/2)$, $2t \in (0, \pi)$, which is exactly the principal range of the arccosine function. Thus, $\cos^{-1}(\cos(2t)) = 2t$. Next, we rewrite the rational part of the integrand:

$$\frac{2x}{(1+x^2)^2} dx = \frac{2 \tan(t)}{(\sec^2(t))^2} \sec^2(t) dt = \frac{2 \sin(t) / \cos(t)}{1 / \cos^2(t)} dt = 2 \sin(t) \cos(t) dt = \sin(2t) dt$$

Substituting everything back into the integral, we get:

$$\int_0^{\pi/2} e^{2t} \sin(2t) dt$$

Now, let $v = 2t$, which implies $dv = 2 dt$, changing the limits to $(0, \pi)$:

$$\frac{1}{2} \int_0^{\pi} e^v \sin(v) dv$$

Using the standard integration formula $\int e^v \sin(v) dv = \frac{e^v}{2}(\sin(v) - \cos(v))$:

$$\frac{1}{2} \left[\frac{e^v}{2}(\sin(v) - \cos(v)) \right]_0^{\pi} = \frac{1}{4} \left(e^{\pi}(0 - (-1)) - e^0(0 - 1) \right) = \frac{e^{\pi} + 1}{4}$$

1.2 Hard Integral

Exponential Root Fractional Integral

Evaluate the definite integral:

$$\int_0^1 \frac{\sqrt{x}}{\sqrt[3]{x} e \sqrt{1-x}} dx$$

Solution: Rewrite the integrand using fractional exponents:

$$I = \int_0^1 x^{1/2}(1-x)^{-1/2}e^{-1/x} dx$$

Apply the substitution $x = \frac{1}{1+u}$. Then $dx = \frac{-1}{(1+u)^2} du$. The boundaries transform as: $x \rightarrow 0 \Rightarrow u \rightarrow \infty$, and $x \rightarrow 1 \Rightarrow u \rightarrow 0$. Substituting these into the integral gives:

$$I = \int_0^\infty \frac{(1+u)^{-1/2}}{e^{1+u} \left(\frac{u}{1+u}\right)^{1/2}} \frac{1}{(1+u)^2} du = e^{-1} \int_0^\infty \frac{u^{-1/2}e^{-u}}{(1+u)^2} du$$

To evaluate this, apply the substitution $u = v^2 \Rightarrow du = 2v dv$:

$$I = 2e^{-1} \int_0^\infty \frac{e^{-v^2}}{(1+v^2)^2} dv$$

Consider the parametric integral $J(a) = \int_0^\infty \frac{e^{-a(1+v^2)}}{1+v^2} dv = e^{-a} \int_0^\infty \frac{e^{-av^2}}{1+v^2} dv$. Differentiating with respect to a yields:

$$J'(a) = -J(a) - e^{-a} \int_0^\infty e^{-av^2} \left(1 - \frac{1}{1+v^2}\right) dv = -J(a) - \frac{\sqrt{\pi}}{2} \frac{e^{-a}}{\sqrt{a}} + J(a) = -\frac{\sqrt{\pi}}{2} \frac{e^{-a}}{\sqrt{a}}$$

Integrating $J'(a)$ from a to ∞ (noting $J(\infty) = 0$) gives:

$$J(a) = \int_a^\infty \frac{\sqrt{\pi}}{2\sqrt{t}} e^{-t} dt = \frac{\pi}{2} \operatorname{erfc}(\sqrt{a})$$

Now define $f(a) = \int_0^\infty \frac{e^{-a(1+v^2)}}{(1+v^2)^2} dv$. Notice that $f'(a) = -J(a)$. Thus:

$$f(1) = \int_1^\infty J(t) dt = \frac{\pi}{2} \int_1^\infty \operatorname{erfc}(\sqrt{t}) dt$$

Using integration by parts ($w = \operatorname{erfc}(\sqrt{t}), dw = -\frac{1}{\sqrt{\pi t}} e^{-t} dt$):

$$\int_1^\infty \operatorname{erfc}(\sqrt{t}) dt = \left[t \operatorname{erfc}(\sqrt{t}) \right]_1^\infty + \frac{1}{\sqrt{\pi}} \int_1^\infty \sqrt{t} e^{-t} dt$$

Substitute $t = x^2$ into the remaining integral to get $2 \int_1^\infty x^2 e^{-x^2} dx$, which integrates by parts to $e^{-1} + \frac{\sqrt{\pi}}{2} \operatorname{erfc}(1)$.

$$\int_1^\infty \operatorname{erfc}(\sqrt{t}) dt = -\operatorname{erfc}(1) + \frac{1}{\sqrt{\pi}} \left(e^{-1} + \frac{\sqrt{\pi}}{2} \operatorname{erfc}(1) \right) = \frac{e^{-1}}{\sqrt{\pi}} - \frac{1}{2} \operatorname{erfc}(1)$$

Substituting this back into $f(1)$, and recognizing our original integral is $I = 2f(1)$:

$$I = 2 \left(\frac{\pi}{2} \left(\frac{e^{-1}}{\sqrt{\pi}} - \frac{1}{2} \operatorname{erfc}(1) \right) \right) = \sqrt{\pi} e^{-1} - \frac{\pi}{2} \operatorname{erfc}(1)$$

Differentials

2.1 Medium Differential

Non-linear Second Order ODE

Solve the differential equation subject to the initial conditions:

$$(y')^2 = -yy''$$

Given $y(0) = 1$ and $y'(0) = 2$, find the value of $y(2)$.

Solution: Rearrange the differential equation:

$$yy'' + (y')^2 = 0$$

Recognize that the left side is the exact derivative of the product yy' :

$$\frac{d}{dx}(yy') = 0$$

Integrating both sides with respect to x gives:

$$yy' = C_1$$

Use the initial conditions $y(0) = 1$ and $y'(0) = 2$ to find the constant C_1 :

$$C_1 = (1)(2) = 2$$

Thus, we have a separable first-order differential equation:

$$y \frac{dy}{dx} = 2 \implies y dy = 2 dx$$

Integrating both sides:

$$\frac{1}{2}y^2 = 2x + C_2 \implies y^2 = 4x + 2C_2$$

Apply the initial condition $y(0) = 1$ again to find C_2 :

$$1^2 = 4(0) + 2C_2 \implies 2C_2 = 1$$

The specific solution is:

$$y(x)^2 = 4x + 1 \implies y(x) = \sqrt{4x + 1}$$

We take the positive root because $y(0) = 1 > 0$. Finally, evaluate at $x = 2$:

$$y(2) = \sqrt{4(2) + 1} = \sqrt{9} = 3$$

2.2 Hard Differential

Riccati Equation

Solve the following differential equation:

$$\frac{dy}{dx} = y^2 - x^2 + 1$$

Given the initial condition $y(0) = \frac{1}{2}$, find the exact value of $y(1)$.

Solution: This is a standard Riccati equation. We use the transformation $y = -\frac{u'}{u}$. The derivative is:

$$y' = -\frac{u''u - (u')^2}{u^2} = -\frac{u''}{u} + y^2$$

Substitute this back into the original ODE:

$$-\frac{u''}{u} + y^2 = y^2 - x^2 + 1 \implies u'' + (1 - x^2)u = 0$$

We look for a solution of the form $u_1(x) = e^{-x^2/2}$. Let's verify it:

$$u_1' = -xe^{-x^2/2}$$

$$u_1'' = -e^{-x^2/2} + x^2e^{-x^2/2} = (x^2 - 1)u_1$$

Substituting this into the linear ODE: $(x^2 - 1)u_1 + (1 - x^2)u_1 = 0$. So, $u_1(x)$ is a valid fundamental solution. Using reduction of order, the second linearly independent solution is:

$$u_2(x) = u_1(x) \int \frac{1}{u_1(t)^2} dt = e^{-x^2/2} \int_0^x e^{t^2} dt$$

The general solution for $u(x)$ is $u(x) = Ae^{-x^2/2} + Be^{-x^2/2} \int_0^x e^{t^2} dt$. Now, construct $y(x) = -u'/u$:

$$y(x) = -\frac{-Axe^{-x^2/2} + B\left(e^{-x^2/2}e^{x^2} - xe^{-x^2/2} \int_0^x e^{t^2} dt\right)}{Ae^{-x^2/2} + Be^{-x^2/2} \int_0^x e^{t^2} dt}$$

Factoring out and canceling $-e^{-x^2/2}$ yields:

$$y(x) = \frac{Ax - Be^{x^2} + Bx \int_0^x e^{t^2} dt}{A + B \int_0^x e^{t^2} dt} = x - \frac{Be^{x^2}}{A + B \int_0^x e^{t^2} dt}$$

Let $K = B/A$. Then $y(x) = x - \frac{Ke^{x^2}}{1 + K \int_0^x e^{t^2} dt}$. Apply the initial condition $y(0) = 1/2$:

$$\frac{1}{2} = 0 - \frac{Ke^0}{1 + 0} \implies K = -\frac{1}{2}$$

Substitute K back into the solution:

$$y(x) = x + \frac{\frac{1}{2}e^{x^2}}{1 - \frac{1}{2} \int_0^x e^{t^2} dt} = x + \frac{e^{x^2}}{2 - \int_0^x e^{t^2} dt}$$

Evaluating at $x = 1$ gives the exact value:

$$y(1) = 1 + \frac{e}{2 - \int_0^1 e^{t^2} dt}$$

Derivatives

3.1 Medium Derivative

Infinite Continued Fraction Derivative

Given $f(x) = \frac{x}{1 + \frac{x}{1 + \dots}}$ for $x > 0$, $f(x) > 0$. Find $f'(1)$.

Solution: Because the function is an infinite continued fraction, it exhibits self-similarity. We can substitute the entire denominator's nested fraction with $f(x)$ itself:

$$f(x) = \frac{x}{1 + f(x)}$$

Multiply both sides by $(1 + f(x))$ to clear the denominator:

$$f(x)(1 + f(x)) = x \implies (f(x))^2 + f(x) - x = 0$$

This is a quadratic equation in terms of $f(x)$. We apply the quadratic formula:

$$f(x) = \frac{-1 \pm \sqrt{1^2 - 4(1)(-x)}}{2} = \frac{-1 \pm \sqrt{1 + 4x}}{2}$$

Since we are given that $f(x) > 0$ for $x > 0$, we must select the positive root:

$$f(x) = \frac{\sqrt{1 + 4x} - 1}{2}$$

Now we find the derivative $f'(x)$ using the chain rule:

$$f'(x) = \frac{1}{2} \cdot \frac{1}{2\sqrt{1 + 4x}} \cdot 4 = \frac{1}{\sqrt{1 + 4x}}$$

Finally, evaluate the derivative at $x = 1$:

$$f'(1) = \frac{1}{\sqrt{1 + 4(1)}} = \frac{1}{\sqrt{5}}$$

3.2 Hard Derivative

Infinite Product Reals

Given the function $f(x) = \prod_{n=2}^{\infty} \left(1 + \frac{x^2}{n^2}\right)^{\frac{(-1)^n}{n}}$. Find $\text{Re}(f'(i))$.

Solution: Take the natural logarithm of both sides to convert the infinite product into an infinite sum:

$$\ln(f(x)) = \sum_{n=2}^{\infty} \frac{(-1)^n}{n} \ln \left(1 + \frac{x^2}{n^2}\right)$$

Differentiate both sides with respect to x using the chain rule:

$$\frac{f'(x)}{f(x)} = \sum_{n=2}^{\infty} \frac{(-1)^n}{n} \left(\frac{1}{1+x^2/n^2} \right) \left(\frac{2x}{n^2} \right) = \sum_{n=2}^{\infty} \frac{2x(-1)^n}{n(n^2+x^2)}$$

Evaluate this logarithmic derivative at $x = i$ (where $x^2 = -1$):

$$\frac{f'(i)}{f(i)} = \sum_{n=2}^{\infty} \frac{2i(-1)^n}{n(n^2-1)} = i \sum_{n=2}^{\infty} \frac{2(-1)^n}{n(n^2-1)}$$

Let the purely real summation be $S = \sum_{n=2}^{\infty} \frac{2(-1)^n}{n(n^2-1)}$. Therefore, $f'(i) = iSf(i)$. Now, let's examine $f(i)$ itself:

$$f(i) = \prod_{n=2}^{\infty} \left(1 - \frac{1}{n^2} \right)^{\frac{(-1)^n}{n}}$$

Since $n \geq 2$, every term in the product $(1 - 1/n^2)$ is strictly positive and real, making $f(i)$ a purely real non-zero number. Because S and $f(i)$ are both purely real, their product $Sf(i)$ is real. Thus, multiplying by i makes $f'(i)$ purely imaginary. Consequently, the real part is zero:

$$\operatorname{Re}(f'(i)) = 0$$

Physics & Engineering

4.1 Electromagnetism

Electrostatic Fluid Pressure

A parallel-plate capacitor is lowered into water in a horizontal position, with water filling the gap of width $d = 1.0$ mm. A constant voltage $V = 1$ kV is applied to the capacitor. Find the water pressure increment Δp (in Pa) in the gap. Assume the relative permittivity of water is $\epsilon = 81$ and the permittivity of free space is $\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$.

Solution: The electric field E established between the parallel plates of the capacitor is uniform and given by:

$$E = \frac{V}{d} = \frac{1000 \text{ V}}{0.001 \text{ m}} = 10^6 \text{ V/m}$$

When a dielectric fluid is drawn into a region of electric field, it experiences an electrostatic body force. By considering the virtual work done in drawing the fluid into the capacitor at constant voltage, the resulting increment in hydrostatic pressure Δp (electrostatic pressure) is equivalent to the change in the energy density of the system compared to free space. This is given by:

$$\Delta p = \frac{1}{2} \epsilon_0 (\epsilon - 1) E^2$$

Substitute the given numerical values into the formula:

$$\Delta p = \frac{1}{2} (8.85 \times 10^{-12}) (81 - 1) (10^6)^2$$

$$\Delta p = \frac{1}{2} (8.85 \times 10^{-12}) (80) (10^{12})$$

The powers of 10 cancel nicely ($10^{-12} \times 10^{12} = 1$):

$$\Delta p = \frac{1}{2} (8.85) (80) = 40 \times 8.85 = 354 \text{ Pa}$$

4.2 Mechanics

Varying Density Sphere Inertia

The densities of two solid spheres A and B of the same radius R vary with radial distance r as $\rho_A(r) = k(r/R)$ and $\rho_B(r) = k(r/R)^5$ respectively, where k is a constant. The moments of inertia of the individual spheres about axes passing through their centres are I_A and I_B , respectively. If $\frac{I_B}{I_A} = \frac{n}{10}$, find the value of n .

Solution: The moment of inertia of a sphere with a spherically symmetric, radially varying density $\rho(r)$ is:

$$I = \frac{8\pi}{3} \int_0^R \rho(r) r^4 dr$$

For sphere A, substitute $\rho_A(r)$:

$$I_A = \frac{8\pi}{3} \int_0^R k \left(\frac{r}{R} \right) r^4 dr = \frac{8\pi k}{3R} \int_0^R r^5 dr = \frac{8\pi k}{3R} \left[\frac{r^6}{6} \right]_0^R = \frac{8\pi k R^5}{18} = \frac{4\pi k R^5}{9}$$

For sphere B, substitute $\rho_B(r)$:

$$I_B = \frac{8\pi}{3} \int_0^R k \left(\frac{r}{R} \right)^5 r^4 dr = \frac{8\pi k}{3R^5} \int_0^R r^9 dr = \frac{8\pi k}{3R^5} \left[\frac{r^{10}}{10} \right]_0^R = \frac{8\pi k R^5}{30} = \frac{4\pi k R^5}{15}$$

Next, find the ratio of their moments of inertia:

$$\frac{I_B}{I_A} = \frac{4\pi k R^5 / 15}{4\pi k R^5 / 9} = \frac{9}{15} = \frac{3}{5}$$

To match the form $\frac{n}{10}$, we multiply the numerator and denominator by 2:

$$\frac{3}{5} = \frac{6}{10}$$

Thus, $n = 6$.

4.3 Quantum Mechanics

Harmonic Oscillator Expectation Value

Consider a quantum harmonic oscillator with ground state $|0\rangle$ and second excited state $|2\rangle$. The ladder operators satisfy $[\hat{a}, \hat{a}^\dagger] = 1$. The position and momentum operators are:

$$\hat{x} = \frac{1}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger), \quad \hat{p} = \frac{i}{\sqrt{2}}(\hat{a}^\dagger - \hat{a})$$

Evaluate the exact numerical value of the ratio:

$$I = \frac{\langle 2 | \hat{x}^2 \hat{p}^2 | 0 \rangle}{\langle 2 | \hat{x}^2 | 0 \rangle}$$

Solution: First, evaluate the denominator. Expand $\hat{x}^2$ in terms of ladder operators:

$$\hat{x}^2 = \frac{1}{2}(\hat{a} + \hat{a}^\dagger)^2 = \frac{1}{2}(\hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}) = \frac{1}{2}(\hat{a}^2 + \hat{a}^{\dagger 2} + 2\hat{n} + 1)$$

When operating on the ground state $|0\rangle$, only the creation operator squared term yields a non-zero overlap with $\langle 2|$:

$$\langle 2 | \hat{x}^2 | 0 \rangle = \frac{1}{2} \langle 2 | \hat{a}^{\dagger 2} | 0 \rangle = \frac{1}{2} \langle 2 | (\sqrt{2}|2\rangle) = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

Next, evaluate the numerator. We first act $\hat{p}^2$ on $|0\rangle$:

$$\hat{p}^2 = -\frac{1}{2}(\hat{a}^\dagger - \hat{a})^2 = -\frac{1}{2}(\hat{a}^{\dagger 2} + \hat{a}^2 - \hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger) = -\frac{1}{2}(\hat{a}^{\dagger 2} + \hat{a}^2 - 2\hat{n} - 1)$$

$$\hat{p}^2|0\rangle = -\frac{1}{2}(\sqrt{2}|2\rangle - |0\rangle) = \frac{1}{2}|0\rangle - \frac{1}{\sqrt{2}}|2\rangle$$

Now, we apply $\langle 2|\hat{x}^2$ to this state:

$$\langle 2|\hat{x}^2\hat{p}^2|0\rangle = \langle 2|\hat{x}^2\left(\frac{1}{2}|0\rangle - \frac{1}{\sqrt{2}}|2\rangle\right) = \frac{1}{2}\langle 2|\hat{x}^2|0\rangle - \frac{1}{\sqrt{2}}\langle 2|\hat{x}^2|2\rangle$$

We already know $\langle 2|\hat{x}^2|0\rangle = \frac{1}{\sqrt{2}}$. We need $\langle 2|\hat{x}^2|2\rangle$:

$$\langle 2|\hat{x}^2|2\rangle = \frac{1}{2}\langle 2|(\hat{a}^2 + \hat{a}^{\dagger 2} + 2\hat{n} + 1)|2\rangle = \frac{1}{2}(0 + 0 + 2(2) + 1) = \frac{5}{2}$$

Substitute this back into the numerator expression:

$$\langle 2|\hat{x}^2\hat{p}^2|0\rangle = \frac{1}{2}\left(\frac{1}{\sqrt{2}}\right) - \frac{1}{\sqrt{2}}\left(\frac{5}{2}\right) = \frac{1-5}{2\sqrt{2}} = -\frac{4}{2\sqrt{2}} = -\sqrt{2}$$

Finally, compute the required ratio I :

$$I = \frac{-\sqrt{2}}{1/\sqrt{2}} = -2$$